

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1148

Acadia 138 kV
February 03, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1148. The Interconnection Customer proposes a 87.1 MW battery energy storage facility interconnecting at the Acadia 138 kV substation in the Pennsylvania area. The request was received on February 13, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 456 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,259,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Tower / Dunsmuir - Adjunct 138 KV	Acadia / Montour Sta 500 KV	193.0	189	10.8
Seney Uppco / Tower 138 KV	Acadia / Motour Station Substation 500 KV	181.8	211	10.8
Hermnas / Seney Uppco 138 KV	Montour Sta / Acadia 500 KV	177.1	269	5.9
Murphy Ck / Sauer 138 KV L	Hermanas 138 kV/1 kV Auto	170.4	170	5.4
Dunsmuir - Adjunct / South Milford Substation 138 KV	Montour Station / Acadia 500 KV	165.9	213	10.8
South Milford / W Marinette Sw Yd Substation 138 KV	Acadia / Montour Station 500 KV Line	147.0	232	10.8
Hemanas 138/1 kV Transformer	Montour Sta / Acadia 500 KV L	126.0	378	5.9
Murphy Crk / Seney Uppco 138 KV	Hermanas 138-1 kV Transformer Bank	122.0	223	5.4
Point Meadows / Susquehanna 138 KV	River St / Vitro Architectural Glass Pa 138 KV Line	116.3	298	11.7
Maunabo 4301 / Pt Meadows 138 KV	Riv St / Vitro Architectural Glass Pa 138 KV	115.7	304	11.7
W Marinette Sw Yd / Maunabo 4301 138 KV	Vitro Architectural Glass Pa / Riv Street 138 KV	114.9	313	11.7
Coronet Substation 500 kV/345 kV Auto	Ft Humbug / Coronet 500 KV	110.6	739	6.0
Walnut Crk 138/1kV TX	Coronet / Ft Humbug 500 KV Line	109.2	911	6.0
Walnut Crk 345/1 kV Autotransformer	Ft Humbug / Coronet 500 KV	109.2	911	6.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Hermanas Substation 345 kV/1 kV XFMR	Acadia / Montour Sta 500 KV	105.1	453	5.9
Jh Warden 500 kV/345 kV TX	Coronet / Westwood Generation Llc 500 KV line	103.9	834	6.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dunsmuir - Adjunct / Tower 138 KV	Rebuild line with higher-capacity conductor	\$12,424,000	14% (\$1,773,000)
Seney Uppco / Tower 138 KV	Rebuild line with higher-capacity conductor	\$3,200,000	14% (\$457,000)
Hermanas / Seney Uppco 138 KV	Rebuild line with higher-capacity conductor	\$4,122,000	16% (\$640,000)
Sauer / Murphy Creek 138 KV	Rebuild line with higher-capacity conductor	\$4,266,000	14% (\$598,000)
Dunsmuir - Adjunct / South Milford Substation 138 KV	Rebuild line with higher-capacity conductor	\$1,551,000	14% (\$221,000)
S Milford / West Marinette Sw Yd 138 KV L	Rebuild line with higher-capacity conductor	\$4,430,000	14% (\$632,000)
Hermanas 138-1 kV Autotransformer	Replace transformer with higher-rated unit	\$4,068,000	16% (\$631,000)
Murphy Crk / Seney Uppco 138 KV	Rebuild line with higher-capacity conductor	\$5,046,000	14% (\$708,000)
Point Meadows / Susquehanna 138 KV	Rebuild line with higher-capacity conductor	\$4,385,000	6% (\$255,000)
Maunabo 4301 / Point Meadows 138 KV	Rebuild line with higher-capacity conductor	\$5,301,000	6% (\$309,000)
W Marinette Sw Yd / Maunabo 301 138 KV	Rebuild line with higher-capacity conductor	\$12,278,000	6% (\$715,000)
Coronet 500 kV/345 kV Auto	Replace transformer with higher-rated unit	\$27,664,000	10% (\$2,811,000)
Walnut Crk 138/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$3,812,000	4% (\$155,000)
Walnut Crk 345/1 kV Autotransformer	Replace transformer with higher-rated unit	\$18,730,000	4% (\$760,000)
Hermanas Substation 345 kV/1 kV XFMR	Replace transformer with higher-rated unit	\$19,572,000	16% (\$3,037,000)
Jh Warden 500 kV/345 kV TX	Replace transformer with higher-rated unit	\$26,914,000	2% (\$557,000)
	Total allocated to project		\$14,259,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.